# A Multimodal Image Database System

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## Abstract

*We demonstrate PBIR$^{MM}$, an integrated system that we have built for conducting multimodal image retrieval. The system combines the strengths of content-based soft annotation (CBSA), multimodal relevance feedback through active learning, and perceptual distance formulation and indexing. PBIR$^{MM}$ supports multimodal query and annotation in any combination of its three basic modes: seed-by-nothing, seed-by-keywords, and seed-by-content. We demonstrate PBIR$^{MM}$ on a couple of very large image sets provided by image vendors and crawled from the Internet.*

## 1 Overview

For a search engine to perform effective searches, it has to comprehend what a user wants. Our demonstration presents a multimodal perception-based image retrieval system (PBIR$^{MM}$), which can capture users' subjective query-concepts thoroughly, and hence achieve high search accuracy. PBIR$^{MM}$ improves over PBIR (our previous demonstration [2]) by the following technologies that we have recently developed:

- Content-based soft annotation,
- Multimodal (textual and perceptual) relevance feedback through active learning, and
- Perceptual distance formulation and indexing.

## 2 Content-based Soft Annotation

Content-based image retrieval supports image searches based on perceptual features, such as color, texture, and shape. However, for most users, articulating a content-based query using these low-level features can be non-intuitive and difficult. Many users prefer to using keywords to conduct searches. We believe that a keyword- and content-based combined approach can benefit from the strengths of these two paradigms. A user can start a query by entering a few keywords. Once some images relevant to the query are found, the image system can use these images' perceptual features, together with their annotation, to perform multimodal query refinement.

Images must be annotated to support such keyword- and content-based combined queries and refinement. In [3] we propose a content-based soft annotation (CBSA) approach to provide images each with multiple semantical labels. The input to CBSA is a training image set, each image in the set is manually annotated with one single semantical label. CBSA propagates these labels to unlabeled images as well as the labeled ones. At the end of the annotation process, each image is annotated with a label-vector, and each label in the vector a confidence factor. For instance, each image in a training set is initially labeled with one of $K$ labels such as forest, tiger, sky, etc. Each image at the end of the CBSA process is annotated with a word vector of $K$ labels. An image label-vector $(forest : 0.1, tiger : 0.9, sky : 0.7, \cdots)$ means that the image is believed to contain semantics of *forest*, *tiger*, and *sky* with $10\%$, $90\%$, and $70\%$ confidence, respectively. When a text-based search is issued with keywords, images are ranked and retrieved based on their combined confidence factors on the matching labels.

The content-based soft annotation algorithm consists of the following three steps:

**1**. Manually labeling a set of training images each with one of the pre-selected $K$ semantical labels.

**2**. Training $K$ classifiers. Based on the labeled instances, we train an ensemble of $K$ BPM (Bayes Point Machine) binary classifiers. Each classifier is responsible for determining the confidence factor for a semantical label.

**3**. Annotating images using the classifiers. Each image is classified by the $K$ classifiers. Each classifier gives each image a confidence factor on the label that the classifier is responsible for predicting. As a result, a $K$-nary vector consisting of $K$-class membership is generated for each image.

## 3 Multimodal Active Learning

As pointed out by [1], automatic annotation may not attain extremely high accuracy at the present state of computer vision and image processing. However, providing images with some reliable semantical labels and then refining these *unconfirmed* labels via relevance feedback is believed an effective approach [14]. CBSA initializes images with a set of semantical words significantly better than chance. Our empirical study shows that even though the initial annotation may not be perfect, CBSA assists a user to quickly find some relevant images via a keyword search. Once some relevant images can be found, query refinement methods such as MEGA [8] and SVM$_{Active}$ [12] can be employed to quickly zoom into the user's query concept.

In [11], we show that the annotation quality can be improved by using user feedback collected from active learning sessions. When a user types in a keyword, say $W$ ($W$ can also be collected from annotated images that are marked by the user as "relevant" to his or her target concept), we select the images which are most difficult for the active learning algorithm to determine their membership to keyword $W$, and use those to so-

licit user feedback. The user simply marks if an image is "relevant" or "irrelevant" to $W$. The active learning algorithm uses the feedback to retrain the binary classifier for keyword $W$, and then adjusts the membership with respect to $W$ in the entire dataset. Our experimental results show this annotation procedure to be very effective in the following two respects:

**1. *Minimum manual effort***. On our testset [11], manually labeling just $10\%$ of images already achieves $90\%$ annotation accuracy for the entire dataset.

**2. *Flexible annotation order***. Users can annotate images one keyword at a time or in a random order, and the annotation quality progresses the same.

## 4 Perceptual Distance Indexing

Similarity is one of the central theoretical constructs in information retrieval. To find objects that are similar to a query object, an information retrieval system must quantify similarity accurately and retrieve similar objects efficiently. We address the central issues of *quantifying perceptual similarity* in [7, 6] and *indexing objects in perceptual (non-metric) spaces* in [4] .

Some important work in the cognitive psychology community has provided ample evidence that similarity is both *dynamic* and *non-metric* [10, 13]. Most recently, Goldstone [9] shows that similarity perception is the *process* that determines the respects (i.e., features, or attributes) for measuring similarity. More precisely, a similarity function for measuring a pair of objects is formulated only after the objects are compared, not before the comparison is made; and the respects for the comparison are *dynamically* activated in this formulation process. Let us use a simple example to explain. Suppose we are asked to name two places that are similar to Los Angeles. Among several possibilities, New York and San Francisco could be two reasonable answers. However, the respects in which Los Angeles is similar to New York differ from those in which Los Angeles is similar to San Francisco. If we use the shared attributes of Los Angeles and New York to compare Los Angeles and San Francisco, the latter pair might not seem similar, and vice versa. Thus, a distance function using a fixed set of respects (e.g., the traditional weighted Minkowski function) cannot capture objects that are similar in different sets of respects.

PBIR$^{MM}$ uses a non-metric distance function, dynamic partial function (DPF), which works significantly better than Minkowski-like functions for measuring perceptual similarity. Unfortunately, since DPF is a non-metric distance function, traditional indexing schemes that require metric-space properties cannot work with DPF. Furthermore, since the *respects* for measuring similarity by DPF are determined in a pairwise fashion, and the feature space in which different object-pairs are compared can be different, coordinate-based indexing structures (e.g., tree-like structures or hash-based structures [5]) cannot work with DPF for conducting similarity searches. We thus employ DynDex (DYnamic Non-metric-space inDEXer) to work with DPF. DynDex uses a statistical approach, which works in two steps. First, DynDex performs pairwise distance-based clustering to group similar objects together. To maximize IO efficiency, each cluster is stored in a sequential file. Second, DynDex models similarity searches as a classification problem. Given a query object $q$, DynDex predicts $q$'s class membership and yields $C$ most probable classes (i.e., clusters). DynDex then finds the objects that have the shortest pairwise distance with respect to $q$, using pairwise respects that are dynamically instantiated. DynDex can achieve very high similarity search accuracy by probing just a small fraction of the dataset. In addition, DynDex scales well with data dimensionality and dataset size, and it holds quite well when features are dynamically weighted to support context-based (or personalized) similarity searches.

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